Senior ML / Code Review — Student Wellbeing Project

> Review date: 2026-06-04 · Reviewer scope: `Scripts/ML/` and `results/` Dataset: 4 participants (Mo, Beatrice, Alvaro, Alessia), participant column = `video_source`.

Executive summary

The **evaluation machinery is genuinely good**: Leave-One-Subject-Out is implemented correctly by `video_source`, preprocessing is fit per-fold, identifiers are excluded from features, the hierarchical classifier is evaluated end-to-end, and unsupervised labels are only attached after clustering. The requirements are almost fully met.

But the headline numbers are not trustworthy, and the reason is upstream of the ML code, in `Scripts/build_training_dataset.py`. The perfect sleepy-vs-stress result is a **data-construction confound**, not a modelling success. This is ranked issue #1 because it changes how every result should be interpreted.

---

1. Requirement checklist

Supervised

Requirement	Status	Notes
Binary normal/abnormal, LOSO by `video_source`	✅ Satisfied	`LeaveOneGroupOut` on `video_source`
Compare LR / SVM / RF / GB(or HistGB)	✅ Satisfied	LR, SVM, RF, GradientBoosting
Binary metrics (acc, bal-acc, macro-F1, P, R, CM, report)	✅ Satisfied	All present, per-fold + overall
Subtype sleepy/stress, LOSO, same models/metrics	✅ Satisfied	— but result is confounded (see Major #1)
Hierarchical: normal→stop, abnormal→subtype	✅ Satisfied	`evaluate_hierarchical_task` correct end-to-end
Hierarchical full metric set + per-fold	✅ Satisfied	weighted-F1, per-class P/R, CM, report all there
Direct 3-class, excludes anomalous.csv	✅ Satisfied	`build_direct_3class_dataset` uses normal+sleepy+stress only
Direct 3-class vs hierarchical comparison	✅ Satisfied	`hierarchical_vs_direct_3class_comparison.csv`

Unsupervised

Requirement	Status	Notes
Binary: KMeans / GMM / Agglomerative, 2 groups, labels post-hoc	✅ Satisfied	—
Anomaly detection trained on normal only	✅ Satisfied	—
LOSO anomaly version (train 3 normals, test held-out)	✅ Satisfied	`run_anomaly_detection_loso` — clean
Binary eval (silhouette, ARI, NMI, mapped acc, bal-acc, macro-F1, CM, report)	🟡 Partial	Metrics yes; **clustering** methods get a cluster-vs-label table but **no confusion-matrix PNG and no classification_report** (only anomaly detectors do). The 3-class script *does* save these — inconsistent.
3-class: KMeans/GMM/Agglo, 3 groups	✅ Satisfied	—
3-class metrics (sil, DB, CH, ARI, NMI, homog, complet, V)	✅ Satisfied	All present
Majority-vote only for interpretation	✅ Satisfied	—
Plots + result CSVs	✅ Satisfied	PCA/t-SNE saved; UMAP optional

Net: essentially all requirements satisfied; only one minor gap (binary clustering confusion-matrix/report parity).

---

2. Major issues (ranked by severity)

🔴 #1 — Sleepy-vs-stress is a protocol confound, not a learned signal *(most important)*

The perfect `1.000` subtype result is the single most suspicious number in the project, and the data confirms why:

- Sleepy HR: mean **51 bpm**, range 27–79. Stress HR: mean **95 bpm**, range 66–126.
- Only **1.9%** of sleepy rows have HR > 65; only **7.8%** of stress rows have HR < 79. The classes are **near-disjoint on a single feature (HR)**, and likewise on RMMSD (sleepy 0.03–0.25 vs stress 0.0009–0.05).

The "stress" recordings were captured under a stress-induction protocol that raised heart rate, and "sleepy" at rest. So the model is detecting **which acquisition condition the heart trace came from**, not

a subtle wellbeing distinction. A perfect LOSO score here is a *red flag*, not a success — it means the task, as constructed, is trivial.

What to do: report this explicitly; run sleepy-vs-stress **with HR/RMSSD removed** (posture+blink only) to show how much signal survives without the confound. The unsupervised ablation already hints at this — extend it to the supervised subtype task.

🔴 #2 — Heart features are joined by row position, from a different session

In `build_training_dataset.py` (`attach_heart`), HR/RMMSD come from separate `Heart_*.csv` files that have **no `window_start` and no `video_source`**, and are attached **by index position** (`iloc[:n]`, trimmed to the shorter length). Consequences:

- HR for window *i* is not time-synchronized with the posture/blink window *i*.
- Because the heart file has no participant key, HR rows are spread across all 4 participants' windows **by position** — the HR assigned to "Mo" windows may not be Mo's. This silently breaks the per-subject integrity that LOSO depends on for the HR channel.
- HR is also the top driver of the abnormal/subtype splits, so this fragility propagates into the most important results.

What to do: re-extract heart features keyed by participant + time window, or at minimum document that HR/RMSSD are *unaligned* and treat them as session-level, not window-level, features. Until then, HR-derived conclusions should be heavily caveated.

🟠 #3 — 4 participants ⇒ LOSO has only 4 folds; high variance

LOSO is the right choice, but n=4 means each fold is one person and estimates swing a lot. The binary RF Mo fold drops to **0.88 acc / 0.81 abnormal-recall**; direct-3-class Mo fold drops to **0.76** for LR/GB. The pooled "overall" numbers (0.957, 0.951) mask this. The interpretation text does report fold ranges (good), but the headline should lead with **mean ± std across the 4 folds**, not the pooled figure, and explicitly state that 4 subjects cannot support generalization claims.

🟠 #4 — Pooled "overall" metrics computed on concatenated fold predictions

`evaluate_flat_task` concatenates all fold predictions and computes one metric set. That's a legitimate "micro-averaged over folds" estimate, but it weights larger participants more and hides per-subject variance. Report it alongside the **macro-over-folds mean ± std**, and use the latter as the primary number. (Per-fold rows are already stored, so this is a reporting change only.)

---

3. Data-leakage audit

All **clean**:

- ✅ Features = the 18 numeric sensor columns only; `window_start`, `video_source`, `label`, `true_label` never enter `X` (`df.loc[:, features]`).
- ✅ Supervised preprocessing fit only on training folds — every model is a `Pipeline` that is `clone()`d and `.fit()` per fold. No global scaler.
- ✅ LOSO anomaly detection fits imputer+scaler+detector **on training-normal only**, then `transform`s the held-out fold. Correct.

- ✅ `anomalous.csv` is excluded from direct 3-class (verified: anomalous = sleepy+stress exactly, 431 rows; direct task rebuilds from the three base files).

One nuance, not strictly leakage: in the **non-LOSO** clustering (`preprocess_full`) the scaler is fit on all rows including the points being evaluated. That's standard and acceptable for transductive clustering, but the silhouette/ARI numbers still pool correlated subjects — worth a sentence acknowledging it.

---

4. Result interpretation

- **Binary 0.957 / subtype 1.000 / hierarchical 0.957** — the binary and hierarchical numbers are *plausible* (posture features like `p_head_tilt_std`/`p_head_tilt_mean` dominate importance, which is physiologically sensible for slumping/drowsiness). The **subtype 1.000 is not credible as a wellbeing result** — see Major #1.
- **Fold stability**: not stable. Alessia/Beatrice ≈ perfect; Mo is consistently the hard fold (0.88 binary, 0.76–0.88 3-class). This is exactly what 4-subject LOSO should reveal and it's good that it does — but it means the pooled headline overstates reliability.
- **Hierarchical ≈ direct (0.957 vs 0.951)**: sensible conclusion that the extra hierarchy buys little; worth stating that the difference is within fold noise, so "hierarchical is better" is not supported.
- **Unsupervised** (GMM binary macro-F1 0.81, 3-class 0.82; LOF LOSO bal-acc 0.78 with **0 anomalous missed but 97 normals flagged**) — the LOF confusion matrix `[[121,97],[0,431]]` shows it basically calls everything anomalous; balanced accuracy is the honest metric there and should be foregrounded over the 0.851 accuracy.
- **Conclusions appropriately limited?** Yes — both summaries repeatedly state "proof-of-concept, 4 participants, more needed." Good scientific hygiene. The one thing missing is naming the **HR confound** as a specific limitation.

---

5. Code quality

Strengths: clean, typed, well-documented, deterministic (`random_state=42`), pipelines used correctly, results organized into clear subfolders, notebooks are thin wrappers that `import` the scripts (so they **cannot drift** — nice choice).

Issues:

- **Heavy duplication across the three scripts**: `validate_features`, `coerce_feature_columns`, `candidate_group_columns`, `infer_group_column`, `_column_is_group_like`, `classification_metrics`, `majority_vote_mapping`, `save_confusion_matrix_png`, `FEATURE_COLUMNS`, `IDENTIFIER_HINTS`, `GROUP_PRIORITY` are copy-pasted. Extract a shared `wellbeing_common.py`. This is ~300 duplicated lines.
- **Inconsistent per-class metric keys**: the supervised script uses `safe_label()` (→ `precision_sleepy`), the unsupervised ones use the raw label in the f-string. Unify.
- **Binary clustering lacks the confusion-matrix PNG + classification_report** that the 3-class script produces — make `evaluate_clustering` symmetric across both scripts (fold this into the shared module).
- **Committed artifacts**: `results/**/.matplotlib/fontlist-v390.json` is a matplotlib cache and shouldn't be in git. Add `results/**/.matplotlib/` to `.gitignore`.
- A couple of unused imports (`Iterable`, `Optional`) in the hierarchical script.

6. Concrete recommendations

ML methodology

1. **Re-report every supervised task as mean ± std over the 4 LOSO folds**, with pooled micro-average as secondary. Add 95% CIs or at least the min/max already computed.
2. **Run the subtype and binary tasks with HR/RMSSD ablated.** This is the single most valuable experiment: it tells you whether posture+blink alone carry the signal, and quantifies how much of the 0.957/1.000 is the HR confound.
3. **Fix or caveat heart-feature alignment** (Major #2). If you can't re-key it, run a sensitivity analysis dropping HR/RMSSD entirely and report both.
4. **Add a permutation / label-shuffle baseline** and a majority-class baseline so reviewers can see the headroom above chance under LOSO.
5. **Nested CV or fixed, justified hyperparameters.** Current models use defaults (`C=1.0`, RF `n_estimators=500`). With 4 folds you can't really tune, so either keep defaults and say so explicitly, or do leave-one-subject-out *inner* CV for tuning. Don't tune on the test fold.
6. **Calibration & uncertainty**: for the binary stage, add `CalibratedClassifierCV` and report Brier score / reliability, and an abstention threshold for the hierarchical router (low-confidence → "uncertain" rather than forcing normal/abnormal).
7. **Subject-aware unsupervised eval**: report clustering ARI/NMI *per held-out subject* too, not just pooled, to match the supervised rigor.

Feature engineering / reporting

8. Rename `RMMSD`→`RMSSD` at the data-build step (the scripts already handle the alias gracefully, but fixing the source avoids the perpetual footnote).
9. Report **per-feature distributions split by class and by participant** — this would have surfaced the HR confound immediately and belongs in the write-up.
10. Add a short **data provenance** section documenting that posture/blink are window-aligned but HR is positionally attached from a separate session.

Additional experiments that would strengthen it

- Posture-only vs blink-only vs HR-only vs all (already done for unsupervised — add it for supervised).
- Group-stratified learning curve: 2 vs 3 training subjects, to show the trend toward (in)stability.
- More participants — the only real fix for generalization.

---

7. Final judgment

Acceptable as a proof-of-concept — yes, with one mandatory caveat. The pipeline design, LOSO discipline, leakage control, and reporting structure are above the level typically seen at this stage; the code is clean and the notebooks stay in sync with the scripts. The requirements are met.

Before any claim of strong generalization, you need:

1. To **explicitly flag and quantify the HR/stress-protocol confound** (and re-run subtype without HR) — otherwise the 1.000 will (rightly) be read as too good to be true.
2. To **fix or formally caveat the positional heart-feature alignment**.
3. **More subjects.** Four-subject LOSO is a sanity check, not evidence of generalization; report mean ± std over folds and stop leading with pooled numbers.
4. Minor: refactor the duplicated code into a shared module and add the missing binary-clustering confusion-matrix/report for parity.

Address #1 and #2 and this moves from "promising prototype with one misleading result" to "honest, defensible proof-of-concept."